

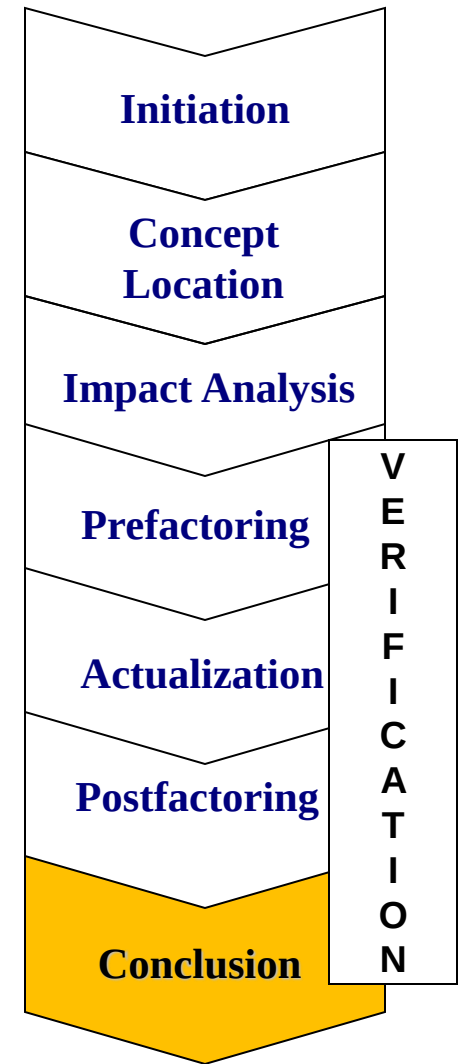
11 Conclusion of software change



The last phase of software change



The activities depend on the specific software process



Steps of conclusion



Commit

programmers return their
updated code to the
configuration
management repository
resolving conflicts



New baseline



New release

New baseline



A thorough testing is required

this testing will guarantee that the new baseline is as bug free as possible

the new baseline represents a progress of the project, not a regression



The baseline testing can take a significant time

often done overnight or over the weekend

a specialized testing team conducts testing

Baseline testing



The frequency depends on the size of the program and required quality



To postpone the baseline testing for too long means that there could be a large accumulation of the bugs

it will make the task of the testing difficult

too frequent baseline testing can represent an unnecessary overhead

Baseline as a deadline



Deadline to commit

time when the testing of the baseline
starts



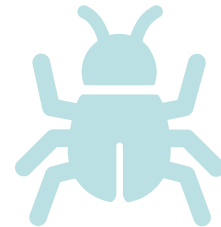
Miss the deadline

submit by the next deadline

- additional work needed for the new baseline
- can be a significant additional work management knows how often a particular programmer missed the deadline
- may require an explanation

Bugs in baseline

- Programmers commit a faulty code
 - bugs are discovered during the baseline testing
- If the bugs are minor, the testing team can still certify the updated code as the new baseline
 - add correction of these bugs into the stack of the bug reports
 - they will be fixed as a part of the future changes.



Broken baseline



More serious bugs

the testing team can reject the buggy commits



Sometimes the whole work done on the new baseline has to be rejected

no new baseline is created

- all work that was done is invalidated or postponed
- a significant cost on large projects

testing team can identify the programmers who committed buggy files

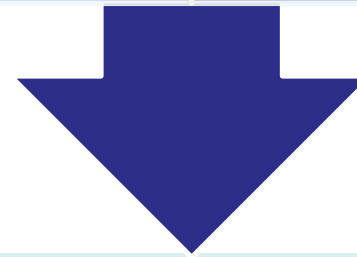
reputation of these programmers suffers

Stakeholder's role

They conduct *acceptance testing*

a functional testing done by the stakeholders

thoroughly tests software functionalities



It gives them information about progress of the project

stakeholders approve software for the release

New release

- From time to time, the programmers release the baseline code to the users.
 - a substantial extra work is required
 - the frequency of the releases is both a technical and a business decision.
- Less frequent large releases + more frequent small releases
 - application AwesomeApp 4.2
 - “versioned model of software lifespan”



Large and small releases

- Download and install large releases
- Small releases are delivered as patches
 - incorporated into the users program by the tool “merge”